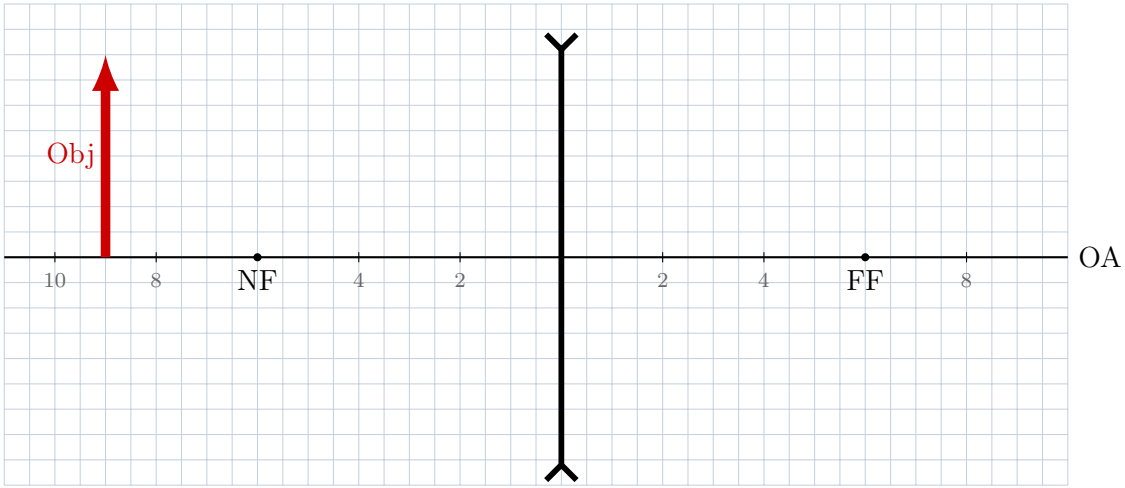
1. A diverging lens has $f = -6$ cm. A 4 cm tall object stands 9 cm in front of it. On the grid, one small square is 0.5 cm.



A. Draw all **three** principal rays for a **diverging** lens. The rays spread apart after the lens, so extend them **back** with dashed lines and mark where they meet.

B. From your drawing, estimate d_i, h_i and M. Is the image real or virtual? Upright or inverted?

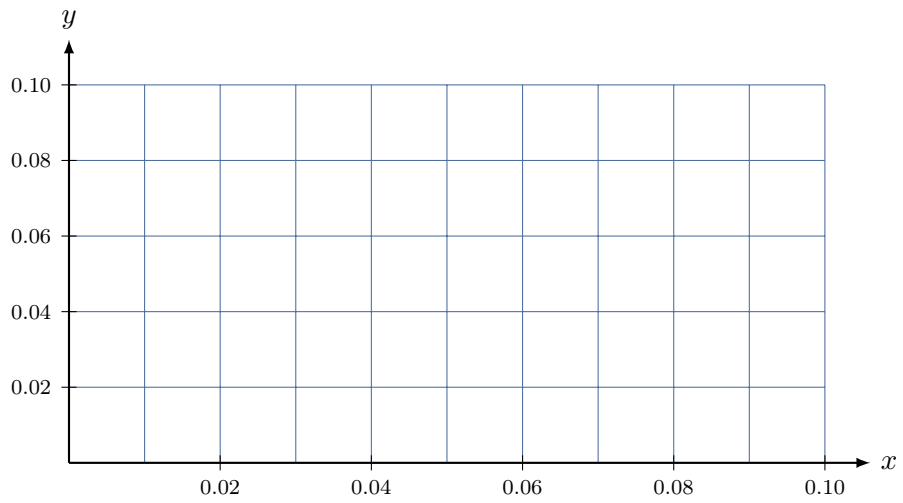
$d_i \approx$ _____ $h_i \approx$ _____ $M \approx$ _____

C. Check with the thin-lens equation and $M = -d_i/d_o$. Do they match your drawing?

2. In lab you measure these image distances for one converging lens:

d_o (cm)	d_i (cm)	$x = 1/d_o$ (cm^{-1})	$y = 1/d_i$ (cm^{-1})
20	30	_____	_____
30	20	_____	_____
60	15	_____	_____

A. Fill in the table. Then plot y against x and draw the best straight line.



B. What are the slope and y-intercept b of your line? Use them to find the focal length of the lens.